

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415253
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Schnitzler; Joshua Michael et al.

Fastener driver

Abstract

A driver for non-magnetic gripping of a socketed fastener head. The driver includes a shank and two cantilever beams. Each cantilever beam has an end and a basal segment adjacent the shank opposite the end. The driver also includes a gap defined by the two cantilever beams in resting positions. The gap extends between the two cantilever beams from an open end toward the shank. The ends of the two cantilever beams are configured to bend into the gap from the resting positions.

Inventors: Schnitzler; Joshua Michael (Lee's Summit, MO), Branson; Donald Conrad (Kansas City, MO)

Applicant: HONEYWELL FEDERAL MANUFACTURING & TECHNOLOGIES, LLC (Kansas City, MO)

Family ID: 1000008818706

Assignee: Honeywell Federal Manufacturing & Technologies, LLC (Kansas City, MO)

Appl. No.: 17/097236

Filed: November 13, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210205964 A1	Jul. 08, 2021

Related U.S. Application Data

continuation parent-doc US 15615639 20170606 US 10864618 child-doc US 17097236

Publication Classification

Int. Cl.: B25B15/00 (20060101); B25B23/10 (20060101); B25G3/28 (20060101); F16B23/00 (20060101)

U.S. Cl.:

CPC B25B15/004 (20130101); B25B23/105 (20130101); B25G3/28 (20130101); F16B23/0007 (20130101); F16B23/0038 (20130101)

Field of Classification Search

CPC: B25B (15/00); B25B (23/10); B25B (23/101); B25B (23/105); B25B (23/108); B25B (23/106); B25G (3/28); B25G (1/005); F16B (23/0007); F16B (23/0038)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2301590	12/1941	Signorelli	81/448	B25B 23/106
2324153	12/1942	Hagness	411/407	B25B 23/106
2562347	12/1950	Young	81/444	B25B 23/106
2657724	12/1952	Shaff	81/444	B25B 23/106
2775913	12/1956	Deliso	81/448	B25B 15/008
2781807	12/1956	Labbee	81/444	F16B 23/0038
4363250	12/1981	Suga	81/453	B25B 23/10
4552039	12/1984	Fisher	294/100	B25B 9/02
6286401	12/2000	Hajianpour	81/453	B25B 23/108
6732426	12/2003	Adams	29/729	B25B 27/10
10864618	12/2019	Schnitzler	N/A	B25B 15/004

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2007021850	12/2006	WO	A61B 17/861

Primary Examiner: Markman; Makena S

Attorney, Agent or Firm: Hovey Williams LLP

Background/Summary

RELATED APPLICATION (1) The current patent application is a continuation patent application which claims priority benefit, with regard to all common subject matter, to identically-titled U.S. patent application Ser. No. 15/615,639, filed Jun. 6, 2017. The earlier-filed patent application is hereby incorporated by reference in its entirety into the current application.

BACKGROUND OF THE INVENTION

1. Field

(1) This invention relates generally to the field of fastener drivers and, in particular, to drivers displaying improved fastener gripping prior to and during installation.

2. Discussion of Related Art

(2) Conventional drivers and bits have long been plagued with fastener-gripping problems. Some outlying changes made to fastener drivers over the past several decades have sought to reduce grip—for example, for use with Phillips-head fasteners—in order to address stripping problems in high torque scenarios. However, most improvements have focused on strengthening the grip a driver has on a fastener, in order to reduce the likelihood of unintended separation during installation. For instance, U.S. Pat. No. 3,253,626 to Stillwagon et al. is directed to a magnetic pick-up fastener driver designed to increase the hold a bit has on a fastener before and during installation.

Additional examples include U.S. Pat. No. 3,298,410 to Noboru and U.S. Pat. No. 7,069,826 to Tilton, each of which is directed to a screw retainer that grips the outer edges of a fastener's head to hold the fastener to the bit before and during installation.

(3) Conventional drivers and bits that rely on attachments to grip the outer edges of fasteners during installation can be too bulky and difficult to operate. Conventional drivers and bits that rely solely on magnetic attraction to hold the fasteners can be too weak and unreliable. A driver with improved grip that addresses one or more of the foregoing concerns is needed.

SUMMARY OF THE INVENTION

(4) The following brief description of embodiments of the invention is provided to indicate the nature of the subject matter disclosed herein. While certain aspects of the present inventive concept are described below, the summary is not intended to limit the scope of the present inventive concept.

(5) The present inventive concept provides, in its simplest form, a solution to the aforementioned issues that includes a driver having at least two flexing cantilever beams for gripping internal surfaces of a fastener socket. Various embodiments of the present inventive concept remedy deficiencies of conventional drivers such as those previously set forth herein.

(6) The aforementioned may be achieved in one aspect of the present inventive concept by providing a driver for gripping a socketed fastener head that includes a shank. The driver also includes two spaced-apart cantilever beams extending from the shank and defining a longitudinal gap between them. Each cantilever beam has an end and a basal segment adjacent the shank. The cantilever beams are configured to narrow the gap between the ends when transitioning from resting positions to assembled positions.

(7) The aforementioned may be achieved in another aspect of the present inventive concept by providing a driver for gripping a socketed fastener head that includes a shank. The driver also includes two spaced-apart cantilever beams extending from the shank and defining a longitudinal gap between them. Each cantilever beam has a basal segment adjacent the shank and an end terminating at a tip. The driver also includes a sliding shim positioned in the gap. The sliding shim and cantilever beams are configured such that moving the sliding shim toward the tips, when the driver is assembled to the socketed fastener head, causes the ends to apply outward pressure on the socketed fastener head.

(8) The aforementioned may be achieved in yet another aspect of the present inventive concept by a method for installing a fastener having a socketed head presenting a top surface and a socket formed in and below the top surface. The method includes providing a driver having two cantilever beams, where each cantilever beam includes an outward-facing surface. The method also includes aligning the driver with a socket of the socketed fastener head and applying a force to the driver causing the cantilever beams to move relative to one another, thereby applying a pressure on the socket.

(9) Additional aspects, advantages, and utilities of the present invention will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the present inventive concept are described in detail below with reference to the attached drawing figures, wherein:
- (2) FIG. 1 is a side perspective view of a driver according to the present inventive concept presenting a hexagonal peripheral profile along a driving end that is bisected through two opposite sides by a gap;
- (3) FIG. 2 is an enlarged front view of the driver of FIG. 1;
- (4) FIG. 3 illustrates multiple fasteners having varied socket types from a side perspective, each fastener being aligned above one or more compatible drivers according to the present inventive concept (illustrated from a front view);
- (5) FIG. 4 is a front view of a driver according to the present inventive concept presenting a hexagonal peripheral profile bisected through two opposite corners by a gap;
- (6) FIG. 5 is a side perspective view of the driver of FIG. 4 just before insertion into a fastener having a hexagonal socket;
- (7) FIG. 6a is a side perspective view of a driver according to the present inventive concept including a sliding shim for enhancing fastener gripping;
- (8) FIG. 6b is an enlarged side view of a portion of the gap and of the sliding shim of the driver of FIG. 6a;
- (9) FIG. 7 is a side view of a portion of a gap and a sliding shim of a driver according to the present inventive concept that includes a track of protrusions for temporarily securing the sliding shim during incremental advancement along the driver;
- (10) FIG. 8a is a side perspective view of a driver according to the present inventive concept including a sliding shim and cantilever beams in resting positions defining a steadily narrowing gap therebetween; and
- (11) FIG. 8b is an enlarged side view of a portion of the gap and of the sliding shim of the driver of FIG. 8a.
- (12) The drawing figures do not limit the present inventive concept to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the present inventive concept.

DETAILED DESCRIPTION OF THE INVENTION

- (13) The following detailed description of the invention references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the present invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.
- (14) In this description, references to “one embodiment”, “an embodiment”, or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment”, “an embodiment”, or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the present technology can include a variety of combinations and/or integrations of the embodiments described herein.
- (15) Turning to FIGS. 1-2, an exemplary driver 50 according to an embodiment of the present inventive concept is illustrated. Driver 50 includes a handle 52, a shank 54 extending from and fixed to the handle 52, and two cantilever beams 56, 58 extending from the shank 54. Cantilever beams 56, 58 respectively include basal segments 60, 62 fixed to or integral with the shank 54, as

well as ends **64**, **66** opposite the basal segments **60**, **62**. Cantilever beams **56**, **58** define a gap **68** between them that extends from a terminus **70** at shank **54** to an open end **72**. Ends **64**, **66** include tips **74**, **76** and outward-facing surfaces **78** and **80**, each outward-facing surface **78**, **80** comprising a plurality of angled surfaces.

(16) The outer peripheries of tips **74**, **76** extend into beveled surfaces. More particularly, between tips **74**, **76** and substantially orthogonal outward-facing surfaces **78**, **80** are, respectively, angled transitional or entry surfaces **82**, **84**. Angled entry surfaces **82**, **84** each comprise a plurality of surfaces angled with respect to each other and configured to facilitate insertion of driver **50** into a socketed fastener head having a hexagonal socket, as explained in more detail below. It is foreseen that angled entry surfaces may be substantially flat, curved, substantially continuous or may take on other configurations without departing from the spirit of the present inventive concept.

(17) Each of ends **64**, **66** also presents a cross-section shaped substantially as an irregular pentagon (See FIG. 2). Viewed together, these cross-sections of ends **64**, **66** present a hexagonal peripheral profile split evenly across two sides by gap **68**.

(18) Turning now to FIG. 3, fastener **86** is illustrated having a hexagonal socket **88** formed in and below a top surface **90**. Socket **88** may include first and second substantially vertical surfaces **92**, **94**, with the first substantially vertical surface **92** being positioned on a generally opposite side of the socket **88** from the second substantially vertical surface **94**. Again considering FIG. 2, the hexagonal peripheral profile presented by the ends **64**, **66** of driver **50** is complementary in shape to the hexagonal socket **88** of fastener **86**. However, in the resting positions of cantilever beams **56**, **58**—i.e., the pre-assembly positions, which in many embodiments means before flexure—the hexagonal peripheral profile presented by ends **64**, **66** is slightly larger in dimension (OD) than the corresponding dimension(s) of the hexagonal socket **88** of fastener **86**. To assemble driver **50** to fastener **86**, cantilever beams **56**, **58** may be flexed or bent toward each other by generally opposing forces having components vectored along F (see FIG. 2) to reduce the span of the ends **64**, **66** of driver **50** along oversized dimension OD and permit full insertion into or assembly with the hexagonal socket **88** of fastener **86**.

(19) Turning to FIG. 4, a closely similar driver **96** is illustrated presenting a complementary peripheral profile to the hexagonal socket **88** of fastener **86**. However, in contrast with the gap **68** of driver **50**, gap **98** of driver **96** splits the hexagonal peripheral profile evenly across two corners to form cross-sections shaped substantially as isosceles trapezoids. It is foreseen that a variety of gap locations—even including locations between asymmetrical cantilever beams—may be employed without departing from the spirit of the present inventive concept. Driver **96** additionally includes a shank, tips **102**, **104**, open end **106**, and angled entry surfaces **108**, **110**.

(20) Turning to FIG. 5, driver **96** also includes outward-facing surfaces **112**, **114** of ends **116**, **118**, respectively. Preferably, all or some of opposing inner edges **113**, **115** of angled entry surfaces **108**, **110** are separated by a shorter span than corresponding opposing upper lip segments **117**, **119** of socket **88**. In this manner, following alignment, driver **96** may be at least partly assembled to fastener **86** by applying downward force having a component directed along FD to move the driver **96** into socket **88**. More particularly, inner edges **113**, **115** of angled entry surfaces **108**, **110** may be aligned with socket **88** and rested just inside of and below the lip segments **117**, **119** of socket **88**. Opposing outer edges **121**, **123** of angled entry surfaces **108**, **110** along over-sized dimension OD may rest just above and outside of corresponding lip segments **117**, **119** of socket **88**.

(21) To complete assembly of driver **96** with socket **88**, additional force having a component along FD may be applied to press ends **116**, **118** deeper into socket **88**. Force directed along FD causing contact with socket **88** may generate reaction forces exerted by the walls and lips of socket **88** on ends **116**, **118**. Of particular importance, reaction forces may be applied by opposing lip segments **117**, **119** of socket **88** onto portions of angled entry surfaces **108**, **110**. These reaction forces tend to move or flex opposing cantilever beams of driver **96** toward each other and into gap **98**, effectively reducing the span of the profile presented by ends **116**, **118** along dimension OD. Inward-facing

surfaces of the ends **116, 118** may be brought in close proximity (leaving at least a portion of the gap **98**) or may abut one another following flexure and assembly.

(22) It is noted that other pinching or squeezing forces having components vectored along **F** may additionally or alternatively be applied to achieve the aforementioned flexure of the ends **116, 118**. For example, fingers may move, or a lever or button may actuate movement of a pinching mechanism mounted to the driver **96** to pinch, the ends **116, 118** together (not shown). It is also noted that some or all of the force required for assembly may be applied to the fastener **86**—for example in a direction opposite to vector **FD** in order to push the lip segments **117, 119** of the fastener **86** past the entry surfaces **108, 110**—without departing from the spirit of the present invention.

(23) The force(s) along **FD**, in the reverse direction of **FD**, and/or along **F** may be steadily and/or increasingly applied until driver **96** is fully assembled with socket **88**.

(24) It is foreseen that drivers according to the present inventive concept may be assembled to fasteners without the use of angled entry surfaces—for example through increased reliance on pinching or squeezing forces to reduce the span of the profile presented by a driver along a dimension **OD**—without departing from the spirit of the present inventive concept.

(25) Once driver **96** is fully seated in socket **88**, installation of the fastener **86** into a receiving surface (for example, into a pilot hole in a piece of sheet metal or board) may begin. As introduced above, during movement and/or torqueing of the assembled driver **96** and fastener **86**, there is risk that the driver **96** may “cam out” or otherwise be unsettled from its seat in socket **88**. This may cause stripping of valuable surfaces defining socket **88** and/or loss of the fastener **86**. However, such stripping and/or cam out is resisted by forces applied by the driver **96**. For instance, ends **116, 118** are resilient, and the flexing required to insert driver **96** into fastener **86** generates reaction forces within the ends **116, 118** that urge them outward toward their original positions. In essence, these reaction forces are applied via outward-facing surfaces **112, 114**, on substantially vertical surfaces and/or lip segments **117, 119** of socket **88**. The force(s) of friction resulting from these reaction forces lead to better grip on the fastener **86** and resistance to cam out and/or stripping.

(26) Returning briefly to FIG. 3, the variety of illustrated common fastener heads are again noted along with the corresponding drivers according to embodiments of the present inventive concept arranged in columns below. The drivers each include two or more cantilever beams and one or more gap(s) between cantilever beams. The outward-facing surfaces of the ends of the cantilever beams are of varying shapes and configurations, and are generally configured to engage at least two opposing vertical surfaces and/or lip segments of the corresponding fastener socket. The gaps between cantilever beams are of varying orientations, widths and/or lengths.

(27) For example, the drivers beneath hexagonal socket fastener **86** each include two opposing outward-facing surfaces respectively located on the ends of two cantilever beams. The outward-facing surfaces of each such driver, when assembled to the fastener **86**, are pressed outward against at least two opposing substantially vertical surfaces and/or lip segments of socket **88**. Upon completion of assembly, the brunt of the gripping forces applied by the driver is felt along opposing surfaces of socket **88** which are in contact with the outward-facing surfaces and which are angled in direct opposition to the outward reaction forces applied by resilient flexing of the ends of the cantilever beams. For instance, the greatest amount of force will be applied by driver **96** on opposing substantially vertical surfaces **92, 94** (and/or on their corresponding opposing lip segments) when driver **96** is assembled to fastener **86** such that gap **98** is parallel to the substantially vertical surfaces **92, 94**.

(28) Similarly, the outward-facing surfaces of the driver illustrated beneath the “slotted” fastener **120** may be pushed into socket **122** formed in and below top surface **128** for assembly, and such outward-facing surfaces may press outward against opposing substantially vertical surfaces **124, 126** and/or their corresponding lip segments; the outward-facing surfaces of the drivers underneath the TORX®-brand fastener **130** may be pushed into socket **132** formed in and below top surface

138 and press outward against at least two opposing substantially vertical surfaces (e.g., **134**, **136**) and/or their corresponding lip segments; the outward-facing surfaces of the drivers underneath the square fastener **140** are pushed into socket **142** formed in and below top surface **148** and press outward against at least two opposing substantially vertical surfaces (e.g., **144**, **146**) and/or their corresponding lip segments; and the outward-facing surfaces of the drivers underneath the Phillips or combination fastener **150** are pushed into socket slot **152** and/or socket slot **154**, in each case formed in and below top surface **156**, and press outward against at least two opposing substantially vertical surfaces of socket slots **152**, **154** and/or their corresponding lip segments. It is also noted that the bottom driver beneath fastener **150** includes third and fourth cantilever beams, enabling flexure of each of the four beams in at least two directions. In each case, reaction forces are applied via outward-facing surfaces on surrounding walls and/or lip segments of the corresponding socket. The force(s) of friction created by these reaction forces at the interfaces between outward-facing surfaces and substantially vertical socket surfaces and/or lip segments may help resist cam out and/or help prevent stripping.

(29) Turning now to FIG. **6a**, a driver **160** according to an embodiment of the present inventive concept is illustrated. Driver **160** includes a handle **162**, a shank **164** extending from and fixed to the handle **162**, and two cantilever beams **168**, **170** extending from the shank **164**. Cantilever beams **168**, **170** respectively include basal segments **172**, **174** fixed to or integral with the shank **164**, as well as ends **176**, **178** opposite the basal segments **172**, **174**. Cantilever beams **168**, **170** define a gap **180** between them that extends from a terminus **182** at shank **164** to an open end **184**. A sliding shim **186** including a shim bar **188** is positioned in the gap **180**. The shim bar **188** preferably comprises material exhibiting a high compression strength, and the shim bar **188**, shank **164** and cantilever beams **168**, **170** may comprise the same steel alloy in some embodiments, though it is foreseen that different materials may be employed without departing from the spirit of the present inventive concept.

(30) The sliding shim **186** includes grips **190**, **192** along opposite sides of the shim bar **188**. The grips **190**, **192** preferably comprise a rubber and/or other textured material providing ample friction and grip to an operator's fingers for actuating manual movement of the sliding shim **186** along gap **180**. It is foreseen that movement of the sliding shim **186** may also or alternatively be actuated using a lever or similar mechanism for multiplying the force applied on the sliding shim **186** without departing from the spirit of the present inventive concept.

(31) As with other drivers described above, assembly of driver **160** to fastener **86** includes flexing ends **176**, **178** toward one another into the gap **180**, and insertion of the ends **176**, **178** into socket **88**. The separation between cantilever beams **168**, **170** following such flexure and insertion varies along the length of the cantilever beams **168**, **170**, the degree of separation being greatest near terminus **182** and smaller near open end **184**. In this configuration, in the manner discussed in more detail above, the ends **176**, **178** apply outward forces on the surrounding socket **88**. To increase the outwardly-directed forces and resistance to cam out and/or stripping—for example in particularly rigorous or increased torque applications—sliding shim **186** may be utilized. The sliding shim **186** may be moved toward the ends **176**, **178** to increase the outwardly-directed forces applied on the socket **88**, and may be moved away from the ends **176**, **178** to reduce such forces (for example to facilitate disassembly of the driver **160** from the fastener **86**).

(32) Turning now to FIG. **6b**, additional detail is illustrated from a side view for a portion of driver **160** including sliding shim **186**. Driver **160** includes a stop **196** comprising a protrusion configured to prevent further movement of the sliding shim **186** along the length of cantilever beams **168**, **170** and thus avoid inadvertent removal of the sliding shim **186** from the open end **184**. It is foreseen that stop **196** may take a variety of shapes and configurations without departing from the spirit of the present inventive concept.

(33) The shim bar **188** also includes a fin **194** along each side of gap **180**. Each fin **194** is essentially a fattened segment of the shim bar **188**—as compared with the portion of shim bar **188**

that resides in gap **180**—situated on a side of gap **180**. Each fin **194** tends to reduce or prevent lateral movement of the shim bar **188** within the gap **180**, instead permitting only movement along the longitudinal axis of driver **160**. It is foreseen that a rail (not shown) or other known system for restricting movement to one axis and/or at a certain point along that axis may be used with the sliding shim without departing from the spirit of the present inventive concept.

(34) Turning to FIG. 7, a variation on the portion of driver **160** illustrated in FIG. **6b** is illustrated. It is initially noted that, with certain exceptions to be discussed in detail below, many of the elements of the driver of FIG. 7 are the same as or very similar to those described in relation to driver **160** above. Therefore, for the sake of brevity and clarity, redundant descriptions will generally be avoided here. Unless otherwise specified, the detailed descriptions of the elements presented above with respect to the driver **160** should therefore be understood to apply at least generally to this embodiment as well.

(35) Cantilever beams **198**, **200** define a gap **202** therebetween. A sliding shim **204** having a shim bar **206** is positioned in gap **202**. A series of aligned protrusions **208** forming a track along cantilever beam **198** extends down the length of gap **202**. Protrusions **208** define a series of cradles **210** therebetween. Shim bar **206** and protrusions **208** are respectively dimensioned to provide clearance for shim bar **206** to slide along the full operable length (i.e., all the way to a stop or other permanent obstacle or to the open end of gap **202**) of resting cantilever beams **198**, **200**.

(36) Upon assembly to a fastener **86**, however, cantilever beams **198**, **200** are pressed together to varying degrees along the longitudinal axis, as described above. Therefore, upon assembly to a fastener **86**, less clearance is provided between the apexes of the protrusions **208** and the cantilever beam **200** the nearer sliding shim **204** gets to the fastener **86**. Successive cradles **210** are increasingly difficult to reach the closer the shim **204** slides to the fastener **86**. Put another way, moving shim **204** from a cradle **210** that is closer to an assembled fastener **86** requires more force than moving the shim **204** from a cradle **210** that is further from the fastener **86**. In turn, more outward force on the socket **88** of the fastener **86** is added by moving the shim **204** toward the fastener **86** to a cradle **210** that is nearer the fastener **86** than to a cradle **210** that is further from the fastener. That is, a greater increase in gripping power is realized by movement toward the fastener **86** between proximate cradles **210** than between distal cradles **210**. It is foreseen that fewer cradles may be defined, for example merely a single proximate “locking” cradle and a single distal “unlocking” cradle, and/or that protrusions of the track may take varying shapes, without departing from the spirit of the present inventive concept.

(37) Turning to FIGS. **8a** and **8b**, a driver **212** according to another embodiment of the present inventive concept is illustrated. It is initially noted that, with certain exceptions to be discussed in detail below, many of the elements of the driver **212** are the same as or very similar to those described in relation to the drivers of embodiments discussed above. Therefore, for the sake of brevity and clarity, redundant descriptions will generally be avoided here. Unless otherwise specified, the detailed descriptions of the elements presented above should therefore be understood to apply at least generally to this embodiment as well.

(38) Driver **212** includes a handle **214**, a shank **216**, and cantilever beams **218**, **220** defining a gap **222** therebetween. Cantilever beams **218**, **220** respectively include ends **224**, **226**. Ends **224**, **226** respectively have tips **228**, **230** and outward-facing surfaces **232**, **234**. A sliding shim **236** is positioned in gap **222**.

(39) Gap **222** of driver **212** has variable width along a longitudinal axis when the cantilever beams **218**, **220** are in resting positions. More particularly, gap **222** gradually narrows as it approaches ends **224**, **226**. The hexagonal profile defined by the cross-sections of ends **224**, **226** may have a minimal or no oversized dimension in the resting position, but instead may be configured to fit snugly within socket **88** of fastener **86** with little or no flexure of cantilever beams **218**, **220** from their resting positions. Therefore, cantilever beams **218**, **220** are flexed to a lesser degree, or may not be flexed at all, to facilitate assembly to fastener **86**. Instead, all or a large part of the added grip

or outward force applied to socket **88** of fastener **86** upon assembly may be generated by moving sliding shim **236** closer to fastener **86** following assembly, generally in the manner described above in connection with the driver **160**. Again, it is foreseen that other features disclosed herein—for example the stop and track of protrusions described in connection with FIGS. **6b** and **7**—may be utilized with the driver **212** without departing from the spirit of the present inventive concept.

(40) It is foreseen that other drivers may be configured according to the present inventive concept to operate with various fastener types and dimensions, and may be designed to maximize the contact surface area for torqueing a particular socket shape and/or to maximize the number of different fasteners with which a driver may be used, without departing from the spirit of the present inventive concept. Drivers according to embodiments of the present inventive concept may be constructed of steel alloy and may be tipped with nickel and/or chromium. The choice of steel alloy may be informed by consideration of an alloy's density, Young's Modulus, yield strength, ultimate tensile strength, elongation at failure and other known physical properties affecting driver design and performance. The choice should, however, also be informed by considering the friction forces required to keep the driver firmly seated within a fastener during a variety of applications, and by the related variables of cantilever beam length, gap width and degree of intended flexure. The choice of material may also be influenced by the coefficients of friction presented by the various materials under consideration, though variation of these properties between commonly used alternatives may in some cases be negligible or overwhelmed by other design considerations. The shank and cantilever beams of a driver may comprise the same or different materials, and may be joined or integral, without departing from the spirit of the present inventive concept. One exemplary embodiment includes a driver designed for installing a standard #2-56 hexagonal socket fastener. The driver may comprise AISISM 8650 Nickel-chromium-molybdenum steel (temper 1000° F.), and may include a gap having a length of one inch (1") and a width of two thousandths of an inch (0.002").

(41) It is foreseen that a wide variety of materials and dimensions may be chosen for drivers by one having ordinary skill in the art following review of this disclosure without departing from the spirit of the present inventive concept. It is also foreseen that known fastener-gripping features—such as outer edge- and/or magnetic-gripping features described above—may supplement features set forth herein without departing from the present inventive concept.

(42) Having now described the features, discoveries and principles of the general inventive concept, the manner in which the general inventive concept is constructed and used, the characteristics of the construction, and advantageous, new and useful results obtained; the new and useful structures, devices, tools, elements, arrangements, parts and combinations, are set forth in the appended claims. It is also to be understood that the following claims are intended to cover all of the generic and specific features of the general inventive concept herein described, and all statements of the scope of the general inventive concept which, as a matter of language, might be said to fall therebetween.

Claims

1. A driver for gripping a socketed fastener head, the driver comprising: a shank; two spaced-apart cantilever beams extending from the shank and defining a longitudinal gap between them, each cantilever beam having an end and a basal segment adjacent the shank, the basal segments of the two cantilever beams being immovable relative to the shank; a sliding shim positioned in the gap, the sliding shim and the cantilever beams being configured so that movement of the sliding shim toward the ends increases outward forces on the cantilever beams for gripping the socketed fastener head; and a stop protrusion extending into the gap and configured to contact and engage with the sliding shim to prevent movement of the sliding shim beyond the stop protrusion in the direction of

the ends.

2. The driver of claim 1, wherein the socketed fastener head presents a top surface, a socket is formed in and below the top surface, and the socket presents a first substantially vertical surface and a second substantially vertical surface across the socket from the first substantially vertical surface, with the driver comprising an outward-facing surface on the end of each of the two cantilever beams, the outward-facing surfaces being oriented, when assembled to the socketed fastener head, to apply forces outward on the first and second substantially vertical surfaces in response to bending of the two cantilever beams from resting positions.
3. The driver of claim 2, wherein the two ends are configured to abut one another when assembled to the fastener.
4. The driver of claim 2, wherein the driver is configured such that at least a portion of the gap between the ends remains when the driver is assembled to the socketed fastener head.
5. The driver of claim 1, wherein the ends of the cantilever beams each present a cross-section having the shape of an isosceles trapezoid.
6. The driver of claim 1, wherein the ends of the cantilever beams each present a cross-section having a rectangular shape.
7. The driver of claim 1, wherein each end of the two cantilever beams includes a tip and a substantially orthogonal outward-facing surface.
8. The driver of claim 1, wherein each end of the two cantilever beams includes a tip, an outward-facing surface, and an angled entry surface between the tip and the outward facing surface configured to facilitate insertion of the driver into the socketed fastener head.
9. The driver of claim 1, further comprising a track of translational protrusions extending inside the gap between the stop protrusion and the shank, the track of translational protrusions defining cradles therebetween for receipt of the sliding shim and incremental movement of the sliding shim with respect to the track of translational protrusions and along the gap.
10. The driver of claim 1, wherein the gap defined by the two cantilever beams in resting positions narrows as it extends from the shank toward the ends.
11. The driver of claim 1, wherein the sliding shim includes fins positioned on opposite sides of the gap.
12. A driver for gripping a socketed fastener head, the driver comprising: a shank; a handle fixed relative to the shank in a longitudinal direction; two spaced-apart cantilever beams fixed relative to the shank in the longitudinal direction, extending from the shank in the longitudinal direction, and defining a longitudinal gap between them, each of the two cantilever beams having a basal segment adjacent the shank and an end terminating at a tip; a sliding shim configured for translational movement relative to the cantilever beams, the sliding shim having a lateral length extending to opposite sides of the cantilever beams, a gap segment between the cantilever beams, and first and second fins positioned on the opposite sides of the cantilever beams, the first and second fins each having a thickness that is greater than a thickness of the gap segment in a direction, the direction being substantially orthogonal to the longitudinal direction and also being substantially orthogonal to the lateral direction, the greater thicknesses of the fins being configured to restrict movement of the sliding shim in the lateral direction; and a stop protrusion extending into the gap and configured to prevent movement of the sliding shim beyond the stop protrusion in the direction of the ends.
13. The driver of claim 12, further comprising a track of translational protrusions extending inside the gap between the stop protrusion and the shank, the track of translational protrusions defining cradles therebetween for receipt of the sliding shim and incremental movement of the sliding shim with respect to the track of translational protrusions and along the gap.
14. The driver of claim 12, wherein the gap defined by the two cantilever beams in resting positions narrows as it extends from the shank toward the open end.
15. A method for installing a fastener having a socketed head presenting a top surface and a socket formed in and below the top surface, the method comprising: providing a driver having two

cantilever beams and a shank, each cantilever beam including an outward-facing surface, an end, and a basal segment adjacent the shank, the basal segments of the two cantilever beams being immovable relative to the shank; aligning the driver with a socket of the socketed fastener head; applying a force to the shank of the driver generally in the direction of the socketed fastener head causing the cantilever beams to move toward one another to permit at least partial insertion into the socket; applying a force to a sliding shim in a gap between the cantilever beams generally in the direction of the socketed fastener head to press the sliding shim in a translational movement, thereby applying a pressure on the socket; and settling the sliding shim against a stop protrusion positioned in the gap between the two cantilever beams, the stop protrusion in the gap preventing movement of the sliding shim beyond the stop protrusion in the direction of the socket.

16. The method of claim 15, further comprising, prior to settling the sliding shim against the stop protrusion, translating the sliding shim over a translational protrusion of a track of translational protrusions extending inside the gap between the stop protrusion and the shank.

17. The method of claim 16, further comprising settling the sliding shim in a cradle defined between the translational protrusion and a next-adjacent translational protrusion of the track of translational protrusions.

18. The method of claim 17, comprising torqueing the driver to install the socketed fastener in a receiving surface followed by applying a force to the sliding shim generally in a direction opposite the socketed fastener head to at least partially reduce the pressure applied by the cantilever beams on the socket.

19. The method of claim 18, wherein the force in the direction opposite the socketed fastener head moves the sliding shim out of the cradle, over the translational protrusion and away from the next-adjacent translational protrusion.
